

Kinova.API.Jaco

R4.0.5

Release notes

Version 1.0.0

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Revisions & Approval

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1 Release notes

1.1 Version 4.0.0

1.1.1 Remove the method ExecuteAPIUnitTests

This method has been removed because all the tests will now be performed from another project.

1.1.2 Exception handling improvement

- Method SetClientConfigurations (CClientConfigurations) now manage ArgumentException when the CClientConfigurations parameter is not valid.
- Method SetClientConfigurations (CClientConfigurations) now manage Exception when a general Exception is caught during the process.

1.1.3 Add a new method – RealSpeedTest

This method is in the class CJacoDiagnosticDataManager and in the class CCommManager. It returns a double that represents the quantity of ms to perform 100 transactions of 128 bytes. 12800 bytes total.

1.2 Version 4.0.1

1.2.1 Security modification

The security is now managed by a DLL called Kinova.DLL.SafeGate. Check the new code example to know the new security protocol when declaring a CJacoArm object.

1.2.2 Modification of CJacoArm constructor

The CJacoArm's constructor now takes a CCypherMessage object instead of a string.

1.2.3 Modification of ValidatePassword method

The method ValidatePassword from the class CPasswordManager now takes a CCypherMessage instead of a string.

1.2.4 Remove the method ReleaseUSBConnection

The API had 2 methods in the process of closing the USB connection. Now everything is confined in the single method CloseConnection of the Class CJacoArm which already exist.

1.2.5 Remove the class CLogsManagement

The class CLogsManagement has been removed and all of its functionalities are in the class CJacoDiagnosticDataManager.

1.2.6 Remove the mapping serialization during profile backup

The mapping serialization has been removed from the profile functionalities.

1.2.7 Serialization/Deserialization method's modification

All serialization and deserialization method of the class CjacoConfigurationManager got their name modified. The serialization's type is no longer specified in the method's name. All the serialization are now binary instead of XML/binary.

1.2.8 New functionality system

All the functionality system has been improve. Does not affect the user but improve API maintenance.

1.2.9 Activity logging

All API activities are now logged in the directory ….\Application Data\Kinova\Products\Jacosoft\API\Data\Logs\Activities

1.2.10 New encryption system

1.3 Version 4.0.2 (compatible only with DSP 4.1.6)

1.3.1 Correct syntax error – GetPeripheralInformationFromJaco

The method's name has been corrected.

1.3.2 New Ubuntu version

From now on, a Ubuntu version of this project will be available.

1.3.3 CPosition class modification

The CPosition class now contains information about the temperature sensors, the CAN errors of each joint and the controlling time.

1.3.4 CClientConfigurations class modification

The CClientConfigurations now accept a list of 20 CUserPosition to customize the advance retracted position.

1.4 Version 4.0.3 (compatible only with DSP 4.1.8 and further)

1.4.1 New functionality – Send trajectory to Jaco

You can now send a trajectory to Jaco and it will execute it. The method SendTrajectoryFunctionnality(CPointsTrajectory PointsTrajectory) will send a serie of point the represents a trajectory and the method GetActualTrajectoryInfo() will show the information about the current trajectory.

1.5 Version 4.0.4 (compatible only with DSP 4.1.8 and further)

1.5.1 Functionality modification – Mapping in the profile

From now on, the functionalities CreateProfile and LoadProfile from the configuration manager take into consideration the control mapping data structure.

1.5.2 New functionality – Get Trajectory FIFO information

This functionality gives you information about the trajectory FIFO which is inside Jaco. That information include: Quantity of point that are still in the FIFO, the usage percentage and the max size of the FIFO. The functionality can be accessed in the control manager.

1.5.3 New functionality – Erase the trajectory FIFO

This functionality erases all trajectories inside the trajectory FIFO. The functionality can be accessed in the control manager.

1.5.4 Communication improvement

All the communication process has been optimized.

1.5.5 Namespace modification

Some functionality has been moved into their right namespace.

1.5.6 New constant - CJacoArm

New constants (VERSION_DSP, VERSION_JOINT1, VERSION_JOINT2, VERSION_JOINT3, VERSION_JOINT4, VERSION_JOINT5, VERSION_JOINT6, VERSION_FINGER1, VERSION_FINGER2, VERSION_FINGER3, VERSION_CAN1, VERSION_CAN2) were added in the class CJacoArm. Those constants can be used as index in the code version's array returned by the functionality GetCodeVersion.

1.5.7 New constant - CVectorEuler

New constants (COORDINATE_X, COORDINATE_Y, COORDINATE_Z, THETA_X, THETA_Y, THETA_Z) were added in the class CVectorEuler. Those constants can be used as index in the Position array and in the Rotation array.

1.5.8 New constant - CVectorAngle

New constants (JOINT_1, JOINT_2, JOINT_3, JOINT_4, JOINT_5, JOINT_6) were added in the class CVectorEuler. Those constants can be used as index in the Angle array.

1.6 Version 4.0.5 (compatible only with DSP 4.2.3)

1.6.1 Modification – Get info about trajectory FIFO

Now, when you ask the information about the trajectory FIFO, you can know the maximum size of the FIFO.

1.6.2 New functionality – GetPositioningInfo

Get information related to positioning. That includes cartesian and angular position.

1.6.3 New functionality – GetCommandInfo

Get information related to the command sent to Jaco.

1.6.4 New functionality – GetForceInfo

Get all information related to the force.

1.6.5 New functionality – GetCurrentInfo

Get all information related to electric current.

1.6.6 New functionality – GetSensorsInfo

Get all information related to the captors. That includes the supply voltage, the current, the average power, the X – Y – X accelerators and the temperatures.

1.6.7 New functionality – SendBasicTrajectory

Send a trajectory without the limitations.

1.6.8 New functionality – GetSingularityVector

Get a vector that shows the nearest singularity.

1.6.9 New functionality – GetGripperInfo.

Get information about Jaco's gripper.